BHARAT SIRMAL — PROJECT KNOWLEDGE BASE

Updated project details for portfolio RAG retrieval | Priority-ordered, self-contained project records

Purpose: Use these facts when answering questions about Bharat Sirmal's projects and internship. Each section is self-contained to support document chunking and semantic search.

Retrieval priority: The priority tags below are editorial suggestions for answer emphasis, not actual vector-database weights. RAG ranking must be implemented in the retrieval pipeline.

1. Explainable Brain Tumor Classification (Priority: High)

Other search terms: Brain tumor detection project; MRI classification; Deep Learning; Explainable AI (XAI); Grad-CAM

Overview: Developed a multi-class brain-tumor MRI classification workflow addressing glioma, meningioma, pituitary tumors, and metastasis.

Processing: Handled axial, coronal, and sagittal MRI views; included clinical features in the described workflow.

Explainability: Used Grad-CAM visualizations to help interpret model predictions.

Technologies: Python, PyTorch, CNN, MRI/NIfTI, Grad-CAM.

Evidence boundary: No validated model accuracy, dataset size, clinical deployment status, or performance metrics were supplied. Do not invent these.

2. Bharat AI — Personal Portfolio Assistant (Priority: High)

Other search terms: Bharat AI; personal AI assistant; portfolio chatbot; RAG; LLM; Gemini; ChromaDB

Purpose: A portfolio assistant that answers questions about Bharat using retrieved information from his personal knowledge base.

Knowledge scope: Profile, education, skills, projects, internship, achievements, and career information.

RAG process: Portfolio chat UI receives a question; the backend retrieves relevant semantic context from vectorized knowledge; Gemini generates a grounded response.

Embedding and storage: The latest resume specifies all-MiniLM-L6-v2 sentence embeddings and ChromaDB for vector storage and retrieval.

User interface: Next.js chat experience with streaming replies and speech-to-text/text-to-speech interaction, as stated in the current resume.

Backend-source note: Original project PDF lists FastAPI. Current resume lists Flask. Verify the actual deployed backend before stating a single implementation as definitive.

Technologies: Python, Gemini API, ChromaDB, Sentence Transformers, Next.js; Flask or FastAPI as described by different source versions.

Evidence boundary: No measured retrieval accuracy, latency, token cost, or production-user counts have been provided.

3. YAOP — Community Management Platform (Priority: Medium)

Other search terms: YAOP; community platform; volunteer system; donation management; event management

Overview: Built a community-focused platform to manage events, volunteers, donations, and administrative record updates.

Implementation: Designed and maintained centralized MongoDB records; validated donation data; organized events and volunteers; built administration workflows and a dashboard with Firebase authentication.

Reported result: Reduced manual administrative effort by approximately 40% and helped minimize input errors, as reported in the supplied project file.

Technologies: Next.js, Express.js, MongoDB, Firebase, Cloudinary.

4. FrameForge — Gaming Product E-Commerce Platform (Priority: Medium)

Other search terms: FrameForge; gaming website; gaming e-commerce; product catalog; MERN; admin dashboard

Overview: Gaming-focused e-commerce website with product browsing, user accounts, shopping cart and order workflows, plus administrator management of users, products, and orders.

Backend and database: Designed MongoDB user, product, and order schemas; engineered RESTful CRUD APIs and role-based access to separate administrative and user records.

Website functionality: Gaming hero section, product listings and details, categories, search/filter, registration and authentication, cart/checkout flow, user dashboards, and admin management.

Development process: Tested and validated APIs with Postman; collaborated in a four-person team using Git.

Technologies: React.js, Express.js, MongoDB, Firebase, REST APIs, Postman, Git.

5. JOJO Store — Full-Stack E-Commerce Platform (Priority: Medium)

Other search terms: JOJO Store; online store; e-commerce; Stripe; Vitest; Firebase

Overview: Developed a storefront and administrator workflows for product search, shopping cart, checkout, inventory, and order management.

Implementation: Integrated authentication, payment processing, cloud-based image upload, responsive UI components, and interaction tests with Vitest.

Technologies: Next.js, TypeScript, Firebase, Stripe, Cloudinary, Vitest.

Source distinction: JOJO Store is listed separately from FrameForge in the current resume; do not merge the two projects without confirmation.

6. Web Development Internship — InAmigos Foundation (Priority: Medium)

Dates: May 2026 to June 2026.

Role: Web Development Intern.

Work: Assisted with website development and maintenance; implemented features and improved site layout, navigation, structure, functionality, visual design, and maintainability.

Technologies: HTML, CSS, JavaScript, React, Node.js, MongoDB.

7. Suggested RAG Answering Rules

- For questions about AI/ML projects, retrieve Brain Tumor Classification and Bharat AI first.
- For web development or full-stack questions, retrieve YAOP, FrameForge, and JOJO Store as applicable.
- For experience or internships, retrieve InAmigos Foundation.
- Keep project names and technologies aligned with their own records; avoid transferring features from one project to another.
- Treat qualitative claims and the YAOP ~40% improvement as source-reported, not independently verified.
- If asked about unsupported measurements, GitHub repository URLs, deployments, or uncertain backend choice, say the knowledge base does not establish the answer.

Document basis: “Bharat_Sirmal_03_Projects.pdf” (original project details) and the latest combined resume available in this conversation (additional projects and updates).